

Topic 6: Topology on Metric Space¹

METRIC SPACE

We studied the metric function last week, which measures the “distance” between elements in a set. Let’s review the definition of a metric function:

Definition 1 (Metric Function and Metric Space)

Let X be a set. If there exists a **metric function** $d : X \times X \rightarrow \mathbb{R}^+$ satisfying the following conditions:

- (1) $d(\mathbf{x}, \mathbf{x}) = 0$ for every $\mathbf{x} \in X$
- (2) $d(\mathbf{x}, \mathbf{y}) > 0$ for every $\mathbf{x} \neq \mathbf{y} \in X$
- (3) $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ for every $\mathbf{x}, \mathbf{y} \in X$
- (4) $d(\mathbf{x}, \mathbf{y}) \leq d(\mathbf{x}, \mathbf{z}) + d(\mathbf{y}, \mathbf{z})$ for every $\mathbf{x}, \mathbf{y}, \mathbf{z} \in X$

then (X, d) is called a **metric space**. We say that d is a **metric** on X .

The last property is also called **triangle inequality**.

For example, the set X can be $X = \mathbb{R}^n$. Then, there are several different metrics d defined on $\mathbb{R}^n$ to form a metric space $(\mathbb{R}^n, d)$. Among the metrics defined on $\mathbb{R}^n$, the most commonly used on is the **Euclidean distance**.

Definition 2 (Euclidean Metric)

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. The **Euclidean distance** of $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}.$$

When $X = \mathbb{R}^n$ and d is the Euclidean metric, we often call this metric space an **Euclidean space**. However, Euclidean metric is not the only metric that can be defined on $\mathbb{R}^n$.

¹Instructors: Yue Yu and Sofia Olguin. This note was prepared for the 2026 UCSB Math Camp for Ph.D. students in economics. It incorporates materials from previous instructors, including Shu-Chen Tsao, ChienHsun Lin, and Sarah Robinson.

Let's see an example to illustrate the motivation of having metrics other than the Euclidean metric. Let's consider a network of individuals, where the "distances" between individuals are all identical. This modeling method motivates the following metric space.

Example 1

Let X be any set, and $\mathbf{x}, \mathbf{y} \in X$. Define

$$d(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

Verify that (X, d) is a metric space.

l^p SPACE

In $\mathbb{R}^n$, each element is $\mathbf{x} = (x_1, x_2, \dots, x_n)$, where each $x_i \in \mathbb{R}$. What if we are interested in the metric for elements look like $\mathbf{x} = (\underbrace{x_1, x_2, \dots}_{\text{infinite of them}})$?

When $\mathbf{x} = (x_1, x_2, \dots, x_n)$, $\sum_{i=1}^n x_i^2$ is finite. However, when $\mathbf{x} = (x_1, x_2, \dots)$, $\sum_{i=1}^{\infty} x_i^2$ is not necessarily finite. We focus on the cases where $\sum_{i=1}^{\infty} x_i^2$ is finite.

Definition 3 (l^2 Space)

Define

$$l^2 = \left\{ (x_1, x_2, \dots) \mid \sum_{i=1}^{\infty} x_i^2 < \infty \right\}$$

And define the metric between $\mathbf{x}, \mathbf{y} \in l^2$:

$$d(\mathbf{x} - \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \left(\sum_{i=1}^{\infty} |x_i - y_i|^2 \right)^{1/2}$$

Then, (l^2, d) is a metric space. We call it a l^2 space.

The l^2 space can be generalized to a l^p space, where $p \geq 1$.

Definition 4 (l^p Space)

Define

$$l^p = \left\{ (x_1, x_2, \dots) \mid \sum_{i=1}^{\infty} |x_i|^p < \infty \right\}$$

And define the metric between $\mathbf{x}, \mathbf{y} \in l^p$:

$$d(\mathbf{x} - \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_p = \left(\sum_{i=1}^{\infty} |x_i - y_i|^p \right)^{1/p}$$

Then, (l^p, d) is a metric space. We call it a l^p space.

Example 2

Verify that l^1 space (i.e., the l^p space with $p = 1$) is a metric space.

OPEN BALL AND CLOSED BALL**Definition 5** (Open Ball)

For a metric space (X, d) , an open ball centered at $x \in X$ with radius $r > 0$ is defined as:

$$B(x, r) = \{y \in X : d(x, y) < r\}$$

Definition 6 (Closed Ball)

For a metric space (X, d) , an closed ball centered at $x \in X$ with radius $r > 0$ is defined as:

$$\overline{B}(x, r) = \{y \in X : d(x, y) \leq r\}$$

An open ball includes all points strictly within the radius but excludes the boundary points where the distance to x is exactly r . A closed ball, on the other hand, includes all points within the radius including those on the boundary.

Note that if we try to visualize an open ball (or a closed ball), it may not look like a "ball" if we are considering a metric space other than the Euclidean space.

Example 3

Let X be $\mathbb{R}$. For any $x, y \in X$, define

$$d(x, y) = |x - y|$$

Draw the open ball

$$B(x = 3, r = 2)$$

Example 4

Let X be $\mathbb{R}^2$. For any $\mathbf{x}, \mathbf{y} \in X$, define

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

Draw the open ball

$$B(x = (0, 0), r = 2)$$

Example 5

Let X be $\mathbb{R}^2$. For any $\mathbf{x}, \mathbf{y} \in X$, define

$$d(\mathbf{x}, \mathbf{y}) = |x_1 - y_1| + |x_2 - y_2|$$

Draw the open ball

$$B(x = (0, 0), r = 2)$$

Example 6

Let X be $\mathbb{R}^2$. For any $\mathbf{x}, \mathbf{y} \in X$, define

$$d(\mathbf{x}, \mathbf{y}) = \max\{|x_1 - y_1|, |x_2 - y_2|\}$$

Draw the open ball

$$B(x = (0, 0), r = 2)$$

Example 7

Let X be any set. For any $\mathbf{x}, \mathbf{y} \in X$, define

$$d(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$$

Write down the sets

$$B(x, r = 1.1)$$

and

$$B(x, r = 0.9)$$

OPEN SET AND CLOSED SET

After having the definition of open ball, we can define an "open set."

Definition 7 (Open Set)

Let (X, d) be a metric space. We call $V \subset X$ an **open set** if:

$$\forall x \in V, \quad \exists r > 0$$

such that

$$B(x, r) \subset V$$

In other words, for every point x in the set V , we draw an open ball centered at x , and the open ball is entirely inside V .

Definition 8 (Closed Set)

Let (X, d) be a metric space. We call $F \subset X$ a **closed** if its complement $X \setminus F$ is an open set.

Theorem 1

Any open ball $B(X, r)$ is an open set, and any closed ball $\overline{B}(X, r)$ is a closed set.

The first is easy to see, but the second requires some proof. However, we will skip the proof in this note.

Example 8

Let X be $\mathbb{R}$. For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}$, define $d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}|$. Check if the following sets are open or closed.

1. $(0, 1)$
2. $(0, \infty)$
3. $[0, \infty)$
4. $[0, 1]$

As the definition of closed set entails the complement set, one might think that a set is either open or closed (but not both). However, this guess is not true.

Example 9

Let X be $\mathbb{R}$. For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}$, define $d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}|$. Verify that the following sets are both open and closed.

1. $\emptyset$
2. $\mathbb{R}$

Let (X, d) be any metric space. Verify that the following sets are both open and closed.

1. $\emptyset$
2. X

Theorem 2

The union of any collection of **open** sets is always an **open** set, and the intersection of any finite number of **open** sets is always an **open** set.

The union of any finite number of **closed** sets is always a **closed** set, and the intersection of any collection of **closed** sets is always a **closed** set.

Note that the intersection of an infinite collection of **open** sets may not be open. Besides, the union of an infinite collection of **closed** sets may not be closed.

Example 10

For the following two questions, let X be $\mathbb{R}$. For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}$, define $d(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}|$.

(1) Consider $I_n = (-\frac{1}{n}, \frac{1}{n})$, $n = 1, 2, \dots$

For any n , is I_n an open set or a closed set (or both)?

Find the intersection of an infinite collection, that is, find $\cap_{n=1}^{\infty} I_n$. Is it an open set or a closed set (or both)?

(2) Consider $J_n = [\frac{1}{n}, 1 - \frac{1}{n}]$, $n = 1, 2, \dots$

For any n , is J_n an open set or a closed set (or both)?

Find the union of an infinite collection, that is, find $\cup_{n=1}^{\infty} J_n$. Is it an open set or a closed set (or both)?

COMPACT SET

For economists, the main goal of learning open ball and open set is to derive the concept of a "compact" set. A compact set is a set that is "small" in a certain way, and various theorems in optimization or statistics rely on assuming that we are analyzing a compact set.

Definition 9 (Open Cover)

A **cover** of a set A , denoted as $\mathcal{F}$, is a collection of sets whose union includes A . In math, it means $A \subset \mathcal{F}$. Or, equivalently,

$$\forall x \in A, \quad \exists V \in \mathcal{F} \text{ such that } x \in V$$

A cover of a set A , denoted as $\mathcal{F}$, is an **open cover** if it is a collection of open sets whose union includes A .

Example 11

Consider $\mathcal{F} = \{(\frac{1}{n}, \frac{2}{n}) | n = 1, 2, \dots\}$.

- (1) Is $\mathcal{F}$ a collection of open sets?
- (2) Is $\mathcal{F}$ an open cover of $(0, 1)$?

Definition 10 (Subcover)

A **subcover** is a smaller collection of sets from the original cover that still covers the entire set A .

Definition 11 (Compact Set)

Let (X, d) be a metric space. $A \subset X$.

A set A is called a compact set if **every** open cover of A has a **finite** subcover. In other words,

$$\forall \text{ open cover } \mathcal{F}, \exists V_1, V_2, \dots, V_n \in \mathcal{F} \text{ such that } A \subset \bigcup_{i=1}^n V_i$$

This means that no matter how you try to cover the set with open sets, you can always find a finite number of those sets that still cover all of A . In other words, compact sets don't have "holes" or "gaps" that require an infinite number of pieces to cover them completely.

Example 12

Use the definition of compact, prove that $(0, 1) \subset \mathbb{R}^1$ is not compact. (Hint: consider the open cover $\mathcal{F} = \{(\frac{1}{n}, \frac{2}{n}) | n = 1, 2, \dots\}$)

One important property of a compact set is that it is a bounded set and a closed set.

Definition 12 (Bounded Set)

Let (X, d) be a metric space. A set $A \subset X$ is called **bounded** if:

$$\exists M > 0 \text{ and } x_0 \in X \text{ such that } d(x, x_0) \leq M \text{ for all } x \in A$$

Example 13

Prove that $(0, 1)$ is bounded.

A bounded set is one where the points don't stretch out infinitely far from each other. You can think of a bounded set as one that can be enclosed within a "box" or "ball" of a certain finite size.

Theorem 3

Let (X, d) be a metric space. If $A \subset X$ is compact, then A is bounded and closed.

Generally speaking, the reverse is not always true for all metric space. However, the reverse is true in an Euclidean space.

Theorem 4: Heine-Borel Theorem

Let (X, d) be a metric space, where $X = \mathbb{R}^n$ and d is the Euclidean metric.

$A \subset X$ is compact if and only if A is closed and bounded.

Heine-Borel Theorem is the most important one of our lecture today. This theorem is often how we verify if a set in $\mathbb{R}^n$ is compact. When we are focusing on an Euclidean space, we don't need to verify the compactness via constructing finite subcovering of any open cover. We only need to verify that the set is bounded and closed.

Example 14

Are the following sets bounded? Are they closed? Are they compact?

1. $[0, 1]$
2. $(0, 1)$
3. $[0, \infty)$
4. $(0, \infty)$

CONVEX SET

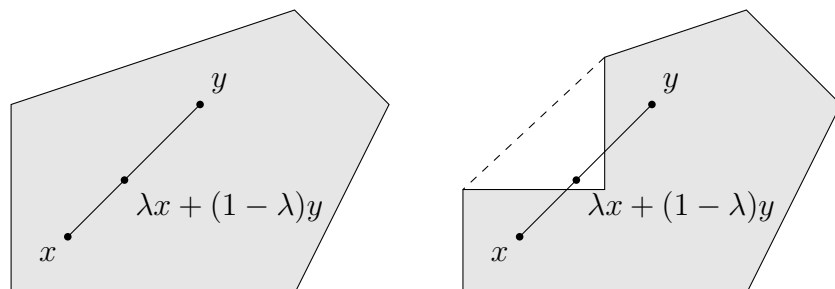
Next, we introduce another important concept widely used in economics and optimization: convex set.

Definition 13

A set $E \subset \mathbb{R}^n$ is a **convex set** if:

$$\forall \mathbf{x}, \mathbf{y} \in E \text{ and } \lambda \in [0, 1], \text{ the point } \lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in E$$

Intuitively, the combination $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$ is a point on the section connecting $\mathbf{x}$ and $\mathbf{y}$. A convex set is a set that contains all sections between two points in the set. In other words, if you can connect two points in the set E , where some parts on the connecting section is not in the set E , then the E is not convex.



Consider any collection of points $p_1, \dots, p_n$. The linear combination

$$\sum_{i=1}^n \alpha_i p_i = \alpha_1 p_1 + \dots + \alpha_n p_n, \quad \text{where } \sum_{i=1}^n \alpha_i = 1, \alpha_i \geq 0 \text{ for every } i$$

is called a **convex combination** of points $p_1, \dots, p_n$.

CONVEX FUNCTION**Definition 14** (Convex/concave functions)

Let $f : X \rightarrow \mathbb{R}$. We say f is a **convex function** if for any $x_0, x_1 \in X$, $\lambda \in [0, 1]$,

$$f(\lambda x_0 + (1 - \lambda)x_1) \leq \lambda f(x_0) + (1 - \lambda)f(x_1).$$

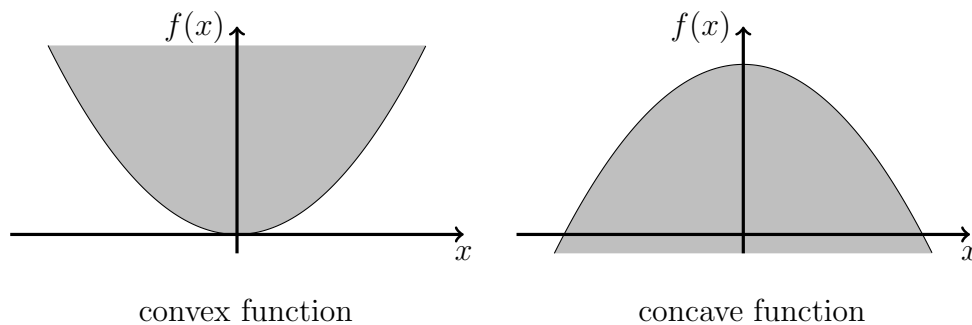
We say f is a **concave function** if for any $x, y \in X$, $\lambda \in [0, 1]$,

$$f(\lambda x_0 + (1 - \lambda)x_1) \geq \lambda f(x_0) + (1 - \lambda)f(x_1).$$

We say f is a **linear function** if f is convex and concave.

If f is a concave function, $-f$ is a convex function. This small trick is frequently used in programming.

Notice very carefully the difference between a convex set and a convex function. The level set (e.g., the contour line for $f(x, y)$) of a convex function is always a convex set.



Here is an useful theorem of using derivatives to verify the convexity of a function $f : \mathbb{R} \rightarrow \mathbb{R}$. We will discuss the multivariate case later.

Theorem 5

For a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is differentiable on an interval I , the the function f is convex if its first derivative is increasing on I . That is,

$$f'(x_1) \leq f'(x_2) \text{ whenever } x_1 < x_2 \in I$$

For a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is twice differentiable on an interval I , the the function f is convex if its second derivative is positive on I . That is,

$$f''(x) > 0 \quad \forall x \in I$$

To verify the concavity of a function f , we can just verify the convexity of $-f$.

Definition 15 (Quasiconvex/quasiconcave functions)

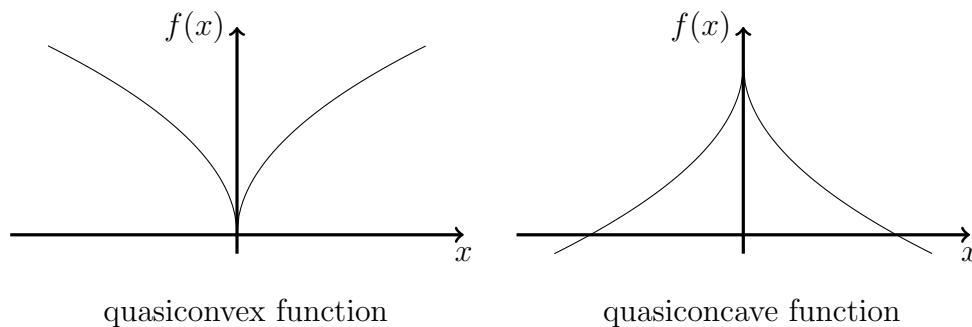
Let $f : X \rightarrow \mathbb{R}$. We say f is a **quasiconvex function** if for any $x_0, x_1 \in X$, $\lambda \in [0, 1]$,

$$f(\lambda x_0 + (1 - \lambda)x_1) \leq \max\{f(x_0), f(x_1)\}.$$

We say f is a **quasiconcave function** if for any $x, y \in X$, $\lambda \in [0, 1]$,

$$f(\lambda x_0 + (1 - \lambda)x_1) \geq \min\{f(x_0), f(x_1)\}.$$

We say f is a **quasilinear function** if f is quasiconvex and quasiconcave.

**Theorem 6**

If f is convex, then it is quasiconvex. If f is concave, then it is quasiconcave.

SOME COMMON INEQUALITIES

Now, we summarize some inequalities commonly used in economics and statistics.

Theorem 7: Jensen's Inequality in $\mathbb{R}^n$

Consider any $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^n$, and any $\lambda_1, \dots, \lambda_n \in [0, 1]$ such that $\sum_{i=1}^n \lambda_i = 1$. This forms a convex combination $\sum_{i=1}^n \lambda_i \mathbf{x}_i$. Suppose f is convex.

The Jensen's inequality states:

$$f\left(\sum_{i=1}^n \lambda_i \mathbf{x}_i\right) \leq \sum_{i=1}^n \lambda_i f(\mathbf{x}_i)$$

Jensen's inequality is basically restating the definition of convex function.

Theorem 8: Cauchy-Schwarz Inequality in $\mathbb{R}^n$

Consider any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, where $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$. The Cauchy-Schwarz inequality states:

$$\left(\sum_{i=1}^n x_i y_i\right)^2 \leq \left(\sum_{i=1}^n x_i^2\right) \left(\sum_{i=1}^n y_i^2\right)$$

For example, in $\mathbb{R}^1$, the Cauchy-Schwarz inequality states $(xy)^2 \leq x^2 y^2$.

Theorem 9: Hölder's Inequality in $\mathbb{R}^n$

Consider any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, where $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$. The Hölder's inequality states:

$$\sum_{i=1}^n |x_i y_i| \leq \left(\sum_{i=1}^n |x_i|^p\right)^{\frac{1}{p}} \left(\sum_{i=1}^n |y_i|^q\right)^{\frac{1}{q}}$$

for all $p, q \in [1, \infty)$ and $\frac{1}{p} + \frac{1}{q} = 1$.

Note that the Hölder's inequality is a generalization of Cauchy-Schwarz inequality.

Example 15

Show that when $p = q = 2$, the Hölder's inequality implies Cauchy-Schwarz inequality. (Hint: there is one more step after letting $p = q = 2$, which involves $|a + b| \leq |a| + |b|$.)

Theorem 10: Minkowski inequality in $\mathbb{R}^n$

Consider any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, where $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$. The Minkowski inequality states:

$$\left(\sum_{i=1}^n |x_i + y_i|^p \right)^{\frac{1}{p}} \leq \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}} + \left(\sum_{i=1}^n |y_i|^p \right)^{\frac{1}{p}}$$

for all $p \in [1, \infty)$.

Example 16

Apply appropriate inequalities to the following questions.

(1) When we have the data $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$, we often evaluate their relationship by calculating the correlation coefficient: $r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$. Prove that $-1 \leq r \leq 1$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

(2) Prove that $\left(\frac{a+b+2c+d}{5}\right)^8 \leq \frac{a^8+b^8+2(c)^8+d^8}{5}$.

(3) In undergraduate statistics, we learned that the variance of X can be written as $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$. Prove that $\mathbb{E}[X^2] - \mathbb{E}[X]^2 \geq 0$ using algebra (i.e., not using the property that $\text{Var}(X) \geq 0$ directly). For simplicity, consider that X is a discrete random variable with some p.d.f. $P(X)$.

From (3), you might wonder if $\mathbb{E}[X^2] - \mathbb{E}[X]^2 \geq 0$ also work for a continuous random variable X . More generally, if the inequalities also work for integrals instead of summation. The answer is yes, and we will revisit these inequalities in the lecture for integrals.